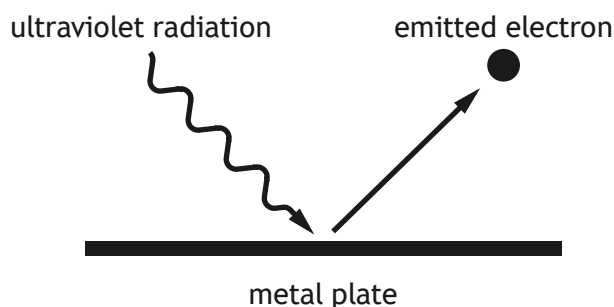


8. Radiation of frequency 1.30×10^{15} Hz from an ultraviolet lamp is incident on a clean metal plate.

This causes electrons to be emitted from the surface of the metal plate.



This is an example of the photoelectric effect.

- (a) Explain why the photoelectric effect provides evidence supporting the particle model of light.

1

- (b) The work function of the metal plate is 5.89×10^{-19} J.

- (i) State what is meant by the term *work function*.

1

- (ii) Calculate the maximum kinetic energy of an electron emitted from the surface of the metal plate.

3

Space for working and answer



8. (b) (continued)

- (iii) The ultraviolet lamp is now moved further away from the metal plate. This reduces the irradiance of the ultraviolet radiation incident on the metal plate.

State what effect, if any, this has on the maximum kinetic energy of an electron emitted from the surface of the metal plate.

You must justify your answer.

2

[Turn over

